

UNIVERSITY OF MPUMALANGA

School of Computing and Mathematical Sciences
Advanced Diploma in Information and Communication Technology
DICT322: Information Systems 322

Transaction Management and Concurrency Control

DISCUSSION QUESTIONS

1. What is a database transaction?
2. What happens when any of the statements in a database transaction fails to complete successfully?
3. A successful database transaction changes a database from consistent state to another. What is a consistent state?
4. What is a database request?
5. Study the SQL code below and explain if it represents a database transaction.

```
SELECT CUST_NUMBER, CUST_BALANCE  
FROM CUSTOMER  
WHERE CUST_NUMBER = 10016;
```

6. How are integrity rules enforced in a database?
7. Discuss properties of a database transaction.
8. What are the SQL statements that provide database transactions support?
9. Examine the following piece of SQL code and answer the questions that follow.

```
INSERT INTO INVOICE  
VALUES (1009, 10016, '18-Jan-2018', 256.99, 20.56, 277.55, 'cred', 0.00, 277.55);  
INSERT INTO LINE  
VALUES (1009, 1, '89-WRE-Q', 1, 256.99, 256.99);  
  
UPDATE PRODUCT  
SET PROD_QOH = PROD_QOH - 1  
WHERE PROD_CODE = '89-WRE-Q';  
  
UPDATE CUSTOMER  
SET CUST_BALANCE = CUST_BALANCE + 277.55  
WHERE CUST_NUMBER = 10016;
```

- a) If the SQL code does not complete execution successfully, will the database remain in a consistent state or not if it was in a consistent state before the execution of the code? Provide an explanation to support your answer.

- b) Explain whether the DBMS will roll back the INSERT INTO INVOICE action that has completed successfully if the rest of the actions of the code do not complete successfully.

10. What is concurrency control and its main objective?
11. Explain why serializability of transactions in a multiuser database environment is important.
12. Explain the data integrity and consistency problems that can occur in a multi-user database environment database environment if transactions are not serialized.
13. Highlight the data integrity and consistency problem in each of the following tables:

TIME	TRANSACTION	STEP	STORED VALUE
1	T1	Read PROD_QOH	35
2	T2	Read PROD_QOH	35
3	T1	PROD_QOH = 35 + 100	
4	T2	PROD_QOH = 35 - 30	
5	T1	Write PROD_QOH (lost update)	135
6	T2	Write PROD_QOH	5

TIME	TRANSACTION	STEP	STORED VALUE
1	T1	Read PROD_QOH	35
2	T1	PROD_QOH = 35 + 100	
3	T1	Write PROD_QOH	135
4	T2	Read PROD_QOH (Read uncommitted data)	135
5	T2	PROD_QOH = 135 - 30	
6	T1	***** ROLLBACK *****	35
7	T2	Write PROD_QOH	105

TIME	TRANSACTION	ACTION	VALUE	TOTAL
1	T1	Read PROD_QOH for PROD_CODE = '11QER/31'	8	8
2	T1	Read PROD_QOH for PROD_CODE = '13-Q2/P2'	32	40
3	T2	Read PROD_QOH for PROD_CODE = '1546-QQ2'	15	
4	T2	PROD_QOH = 15 + 10		
5	T2	Write PROD_QOH for PROD_CODE = '1546-QQ2'	25	
6	T1	Read PROD_QOH for PROD_CODE = '1546-QQ2'	25	(After) 65
7	T1	Read PROD_QOH for PROD_CODE = '1558-QW1'	23	(Before) 88
8	T2	Read PROD_QOH for PROD_CODE = '1558-QW1'	23	
9	T2	PROD_QOH = 23 - 10		
10	T2	Write PROD_QOH for PROD_CODE = '1558-QW1'	13	
11	T2	***** COMMIT *****		
12	T1	Read PROD_QOH for PROD_CODE = '2232-QTY'	8	96

14. What is a scheduler and explain its purpose?
15. What are the concurrency control algorithms used by the scheduler to schedule transactions?
16. Explain the following lock granularity:
- Database-level lock
 - Table-level lock
 - Page-level lock
 - Row-level lock
17. Highlight any challenges that may be experienced when the scheduler uses locks and how are the challenges managed by the scheduler?
18. Explain the two schemes used to decide which transaction is rolled back and which continues executing if there is a conflict between the transactions.

19. Wait/Die and Wound/Wait Schemes are the control schemes used by some concurrency control algorithms. The table below shows 2 transactions, T1 and T2 with their corresponding times of obtaining locks on a particular data item. Contrast how these 2 schemes work by completing the table.

Transaction requesting lock	Transaction owning lock	Wound/wait scheme	Wait/die scheme
T1 (110012)	T2 (110011)		
T2 (110011)	T1 (110012)		

20. What is database recovery and why is it important?
21. Explain the 2 main important concepts that affect the recovery process.
22. Illustrate how the Wait/Die and Wound/Wait Schemes work by completing Table 4, given transactions T1 and T2 with their corresponding times of obtaining locks on a particular data item.

Transaction requesting lock	Transaction owning lock	Wait/die scheme	Wound/wait scheme
T1 (8789)	T2 (2545)		
T2 (2545)	T1 (8789)		

23. Study Figure 1 and answer the questions that follow.

TIME	TRANSACTION	REPLY	LOCK STATUS	
			DATA X	DATA Y
0			Unlocked	Unlocked
1	T1:LOCK(X)	OK	Locked	Unlocked
2	T2:LOCK(Y)	OK	Locked	Locked
3	T1:LOCK(Y)	WAIT	Locked	Locked
4	T2:LOCK(X)	WAIT	Locked	Locked
5	T1:LOCK(Y)	WAIT	Locked	Locked
6	T2:LOCK(X)	WAIT	Locked	Locked
7	T1:LOCK(Y)	WAIT	Locked	Locked
8	T2:LOCK(X)	WAIT	Locked	Locked
9	T1:LOCK(Y)	WAIT	Locked	Locked
...				
...				
...				
...				

- a) Comment on the condition shown in the figure.

b) Describe how the condition mentioned in question (a) is created.

c) Suggest, with an explanation, possible control technique for the condition in the figure.

24. Examine the following Tables 2, 3 and 4, showing 2 transactions taking place concurrently with no concurrency control. Describe the concurrency control problem that each table represents. After the description of each problem, refer to the transactions and the data in the table to specifically highlight the problem.

Time	T ₁	T ₂	bal _X
t ₁	read(bal _X)		12
t ₂	bal _X = bal _X + 12	read(bal _X)	12
t ₃	write(bal _X)	bal _X = bal _X - 3	24
t ₄		write(bal _X)	9

Time	T ₁	T ₂	bal _X	Sum
t ₁	sum=0		17	0
t ₂	read(bal _X)	read(bal _X)	17	0
t ₃	sum = sum + bal _X	bal _X = bal _X - 7	17	17
t ₄		write(bal _X)	10	17

Time	T ₁	T ₂	bal _X
t ₁	read(bal _X)		23
t ₂	bal _X = bal _X + 27		23
t ₃	write(bal _X)		50
t ₄		read(bal _X)	50
t ₅	rollback	bal _X = bal _X - 27	23

25. Examine the following piece of SQL code and answer the questions that follow.

UPDATE PRODUCT

SET PROD_QOH = PROD_QOH - 1

WHERE PROD_CODE = '89-WRE-Q';

UPDATE CUSTOMER

SET CUST_BALANCE = CUST_BALANCE + 277.55

WHERE CUST_NUMBER = 10016;

- a) Identify the database requests from the piece of code.
- b) Will the database still be in a consistent state if the SQL statements above do not complete unsuccessfully?
- c) Write a database transaction from the SQL code.
- d) Explain whether the DBMS will roll back the UPDATE PRODUCT action that has completed successfully if the UPDATE CUSTOMER action does not complete successfully.